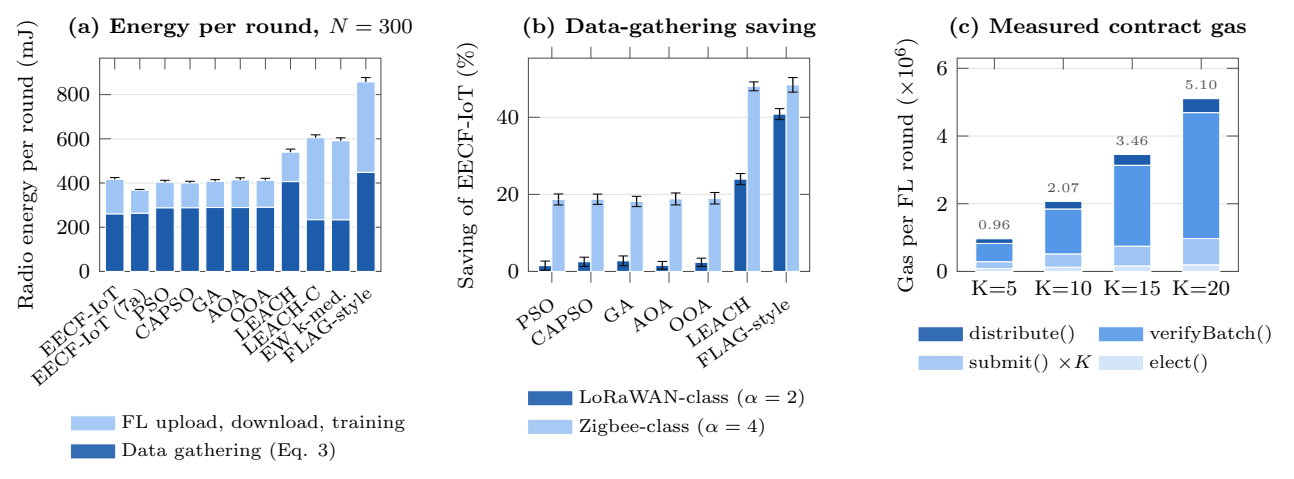

(a) Energy per round, $N = 300$
Radio energy per round (mJ)
0
200
400
600
800
EECF-IoT
EECF-IoT (7a)
PSO
CAPSO
GA
AOA
OOA
LEACH
LEACH-C
EW. k-med.
FLAG-style
FL upload, download, training
Data gathering (Eq. 3)
(b) Data-gathering saving
Saving of EECF-IoT (%)
0
20
40
PSO
CAPSO
GA
AOA
OOA
LEACH
FLAG-style
LoRaWAN-class ($\alpha = 2$)
Zigbee-class ($\alpha = 4$)
(c) Measured contract gas
Gas per FL round ($\times 10^6$)
0
2
4
6
0.96
2.07
3.46
5.10
K=5
K=10
K=15
K=20
distribute()
verifyBatch()
submit() $\times K$
elect()